AI101 Mini-Project · Team 41

Basketball Match Winner Predictor

Predicts which team wins a match using logistic regression

Repo github.com/Mayur-Ukey-358/AI101-Project

MU
Mayur Ukey
Documentation lead
2026MCB1579

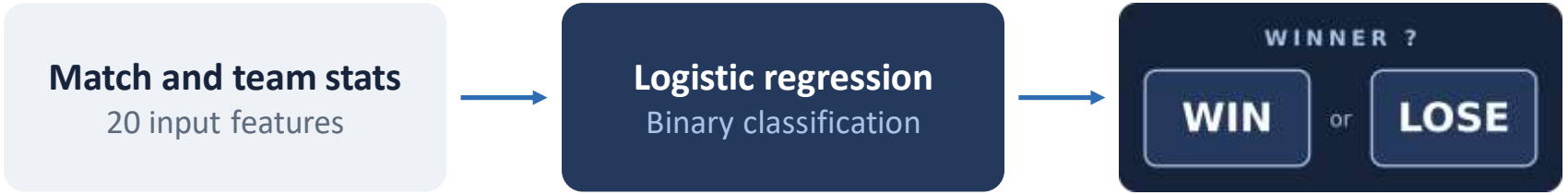
RP
Rushabh Pethad
Execution lead
2026EEB1403

PS
Pratyaksh Soni
PPT lead
2026ESB1488

AU
Abhijith Uttaravelli
Video lead
2026AIB1089



Problem and data



WHAT AND WHY

Predict which of two teams wins a match from pre-match team statistics: ratings, recent form, rest and home advantage. It shows which factors drive wins and how far a simple model can go.

10,000 rows | **22** columns | **51% / 49%** class 0 / class 1

DATASET SOURCE

Kaggle: Basketball Tournament Dataset. Simulated, so results show the method works, not real games.

COLUMNS

Per team (A_ and B_): Off, Def and Net rating, Star Player, Chemistry, Injury, Last5 WinPct, Momentum, Rest. Match: Home_Team, Tournament_Stage. Dropped Match_ID.

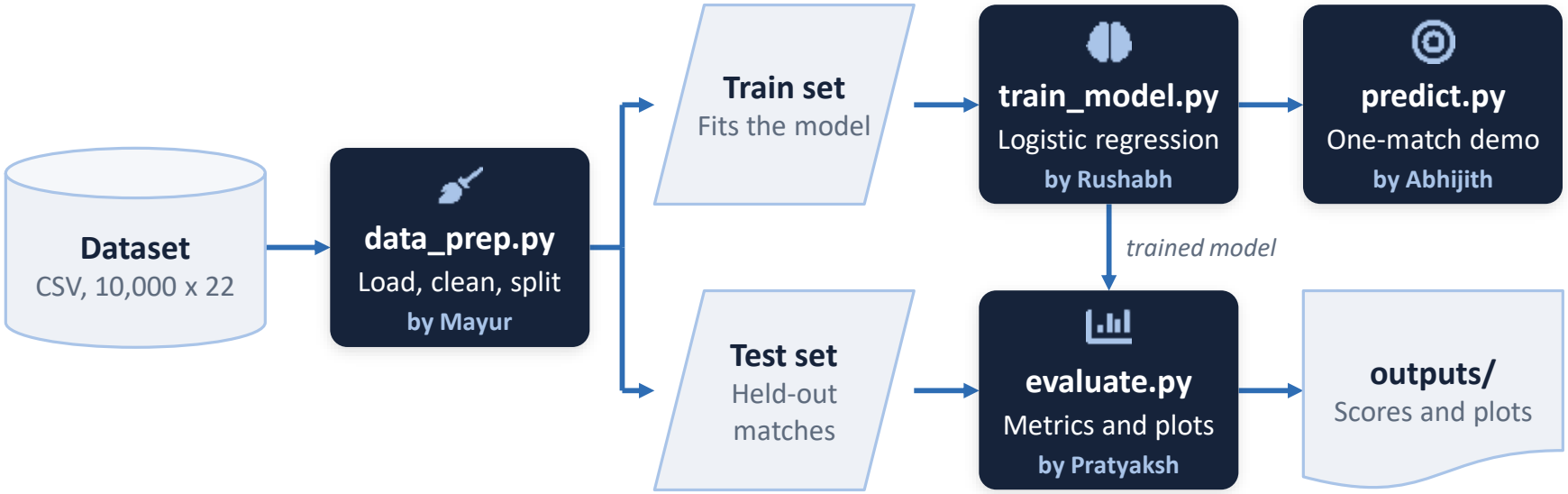
TARGET

Winner: 1 = Team A wins, 0 = Team B wins

Approach

`main.py` runs the whole pipeline

DATA | PREPARE | SPLIT | MODEL | RESULT



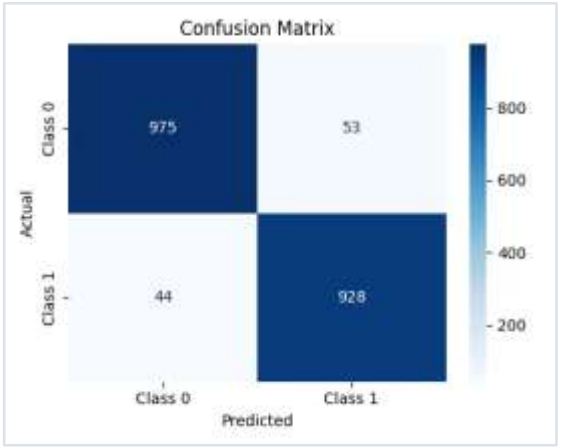
Libraries

pandas | numpy | matplotlib | scikit-learn | joblib | Python 3.10+

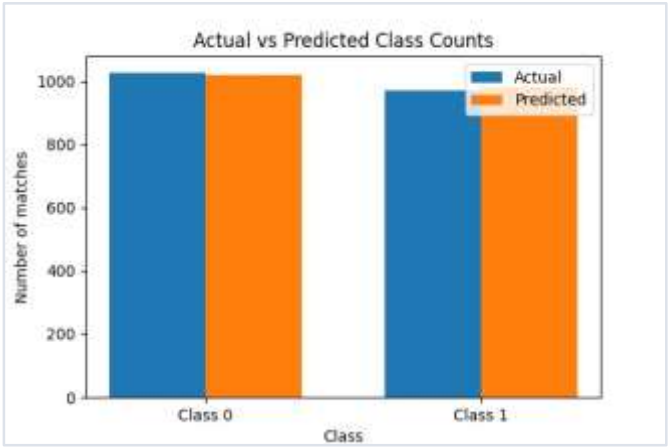
Results

95.2%	0.946	0.955	0.95
Accuracy	Precision	Recall	F1-score

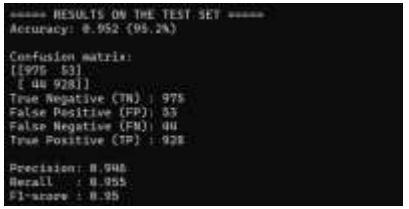
CONFUSION MATRIX



ACTUAL VS PREDICTED



DEMO RUN



Test accuracy 95.2% against 51.4% for always guessing. Training 94.8% and test 95.2%: no overfitting.

Learnings

WHAT WORKED

- Scaling plus logistic regression reaches 95.2%, far above 51.4% for guessing
- Training 94.8% and test 95.2%: no overfitting
- Errors are balanced: 53 false positives, 44 false negatives

WHAT DIDN'T / LIMITS

- Data is simulated, not real games
- Net Rating = Off - Def, so those features overlap
- Only a linear model, no tuning or other models tried

WHO DID WHAT

- **Mayur:** data_prep.py, README
- **Rushabh:** train_model.py, main.py, requirements.txt
- **Pratyaksh:** evaluate.py, slides
- **Abhijith:** predict.py, video

AI-use disclosure Claude (Anthropic) helped structure and debug the code and draft the README and slides. We ran and tested every file ourselves, and each member can explain their part. All results come from running our code on the Kaggle dataset.